

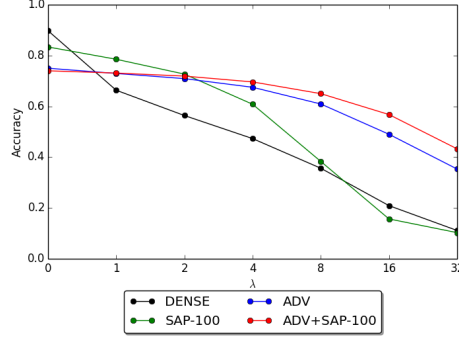


Figure 3: Accuracy plots of the dense, SAP-100, ADV and ADV+SAP-100 models, against adversarial attacks, using MC sampling, with a variety of perturbation strengths, λ .

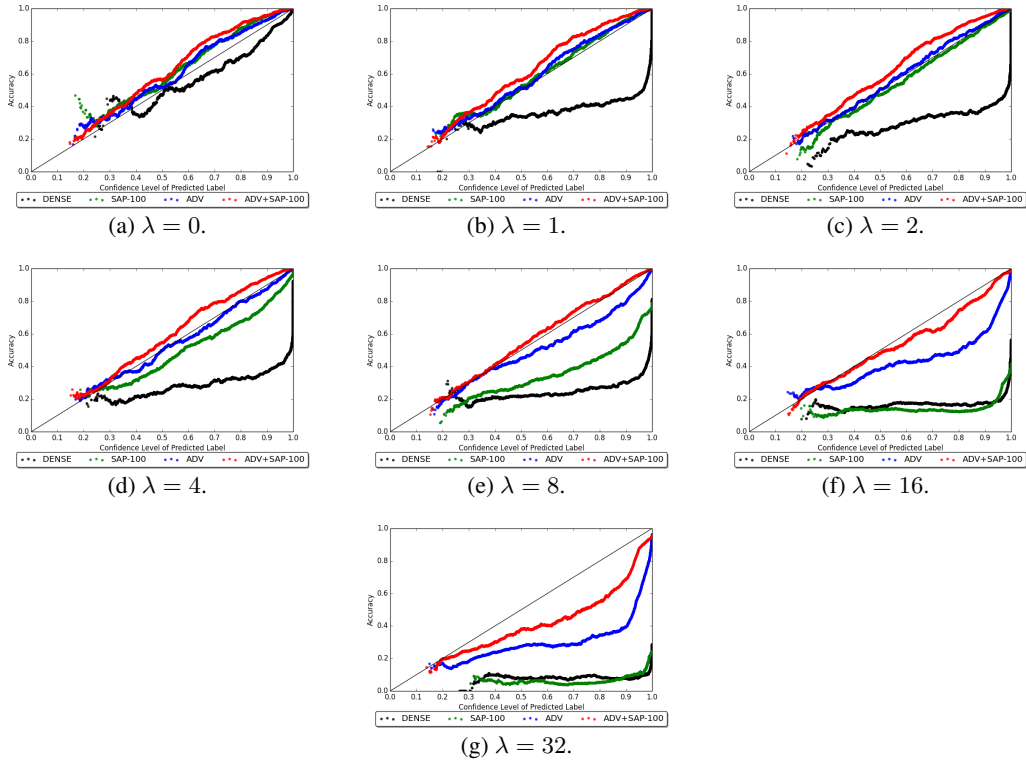


Figure 4: Calibration plots of the dense, SAP-100, ADV and ADV+SAP-100 models, against adversarial attacks, using MC sampling, with a variety of different perturbation strengths, λ . These plots show the relation between the confidence level of the model's output and its accuracy.

the no perturbation case. But for $\lambda \neq 0$, the relative increase in rewards is positive (except for $\lambda = 1$ in the BattleZone game), and is very high in some cases (3425.9% for $\lambda = 1$ for the Bowling game).

4.3 ADDITIONAL BASELINES

In addition to experimenting with SAP, dropout, and adversarial training, we conducted extensive experiments with other methods for introducing stochasticity into a neural network. These techniques included 0-mean Gaussian noise added to weights (RNW), 1-mean multiplicative Gaussian noise for the weights (RSW), and corresponding additive (RNA) and multiplicative (RSA) noise added to

Table 1: Relative percentage increase in rewards gained for SAP-100 compared to original model while playing different Atari games.

λ	Assault	Asterix	BankHeist	BattleZone	BeamRider	Bowling
0	-12.2%	-33.4%	-59.2%	-65.8%	-15.8%	-4.5%
1	10.4%	13.3%	131.7%	-22.0%	164.5%	3425.9%
2	9.8%	20.8%	204.8%	110.1%	92.3%	
4	12.4%	14.0%	1760.0%	202.6%		
8	16.6%	7.4%	60.9%	134.8%		

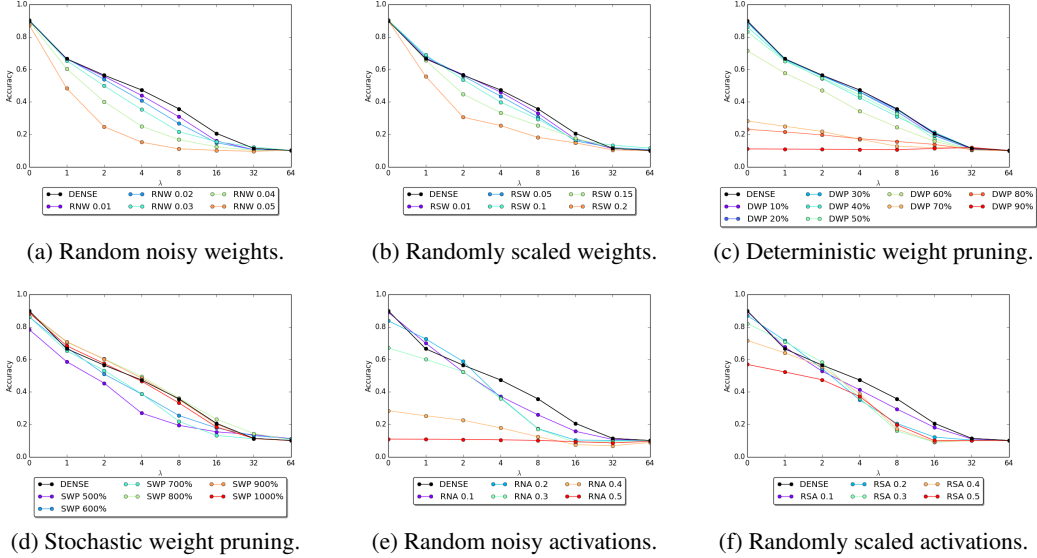


Figure 5: Robustness of different pruning and noisifying strategies against their respective adversarial attacks (MC sampling used to estimate gradients of stochastic models).

the activations. We describe each method in detail in Appendix B. Each of these models performs worse than the dense baseline at most levels of perturbation and none matches the performance of SAP. Precisely why SAP works while other methods introducing stochasticity do not, remains an open question that we continue to explore in future work.

4.4 SAP ATTACKS WITH VARYING NUMBERS OF MC SAMPLES

In the previous experiments the SAP adversary used 100 MC samples to estimate the gradient. Additionally, we compared the performance of SAP-100 against various attacks, these include the standard attack calculated based on the dense model and those generated on SAP-100 by estimating the gradient with various numbers of MC samples. We see that if the adversary uses the dense model to generate adversarial examples, SAP-100 model’s accuracy decreases. Additionally, if the adversary uses the SAP-100 model to generate adversarial examples, greater numbers of MC samples lower the accuracy more. Still, even with 1000 MC samples, for low amounts of perturbation ($\lambda = 1$ and 2), SAP-100 retains higher accuracy than the dense model.

Computing a single backward pass of the SAP-100 model for 512 examples takes ~ 20 seconds on 8 GPUs. Using 100 and 1000 MC samples would take ~ 0.6 and ~ 6 hours respectively.

4.5 ITERATIVE ADVERSARIAL ATTACK

A more sophisticated technique for producing adversarial perturbations (than FGSM) is to apply multiple and smaller updates to the input in the direction of the local sign-gradients. This can be done by taking small steps of size $k \leq \lambda$ in the direction of the sign-gradient at the updated point and

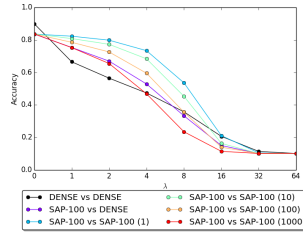


Figure 6: Accuracy plots of adversarial attacks, with different perturbation strengths, λ . The legend shows defender vs. adversary models (used for gradient computation), and the number of MC samples used to estimate the gradient.

repeating the procedure $\lceil \frac{\lambda}{k} \rceil$ times (Kurakin et al., 2016) as follows

$$x^0 = x, \quad x^t = \text{clip}_{x,\lambda} \left(x^{t-1} + k \text{sign}(\mathcal{J}(\theta, x^{t-1}, y)) \right),$$

where function $\text{clip}_{x,\lambda}$ is a projection into a L_∞ -ball of radius λ centered at x , and also into the hyper-cube of image space (each pixel is clipped to the range of $[0, 255]$). The dense and SAP-100 models are tested against this adversarial attack, with $k = 1.0$ (Fig. 1c). The accuracies of the dense model at $\lambda = 0, 1, 2$ and 4 are 89.8%, 66.3%, 50.1% and 31.0% respectively. The accuracies of the SAP-100 model against attacks computed on the same model (with 10 MC samples taken at each step to estimate the gradient) are 83.3%, 82.0%, 80.2% and 76.7%, for $\lambda = 0, 1, 2, 4$ respectively. The SAP-100 model provides accuracies of 83.3%, 75.2%, 67.0% and 50.8%, against attacks computed on the dense model, with the perturbations $\lambda = 0, 1, 2, 4$ respectively. Iterative attacks on the SAP models are much more expensive to compute and noisier than iterative attacks on dense models. This is why the adversarial attack computed on the dense model results in lower accuracies on the SAP-100 model than the adversarial attack computed on the SAP-100 model itself.

5 RELATED WORK

Robustness to adversarial attack has recently emerged as a serious topic in machine learning (Goodfellow et al., 2014; Kurakin et al., 2016; Papernot & McDaniel, 2016; Tramèr et al., 2017b; Fawzi et al., 2018). Goodfellow et al. (2014) introduced FGSM. Kurakin et al. (2016) proposed an iterative method where FGSM is used for smaller step sizes, which leads to a better approximation of the gradient. Papernot et al. (2017) observed that adversarial examples could be transferred to other models as well. Madry et al. (2017) introduce adding random noise to the image and then using the FGSM method to come up with adversarial examples.

Being robust against adversarial examples has primarily focused on training on the adversarial examples. Goodfellow et al. (2014) use FGSM to inject adversarial examples into their training dataset. Madry et al. (2017) use an iterative FGSM approach to create adversarial examples to train on. Tramèr et al. (2017a) introduced an ensemble adversarial training method of training on the adversarial examples created on the model itself and an ensemble of other pre-trained models. These works have been successful, achieving only a small drop in accuracy from the clean and adversarially generated data. Nayebi & Ganguli (2017) proposes a method to produce a smooth input-output mapping by using saturating activation functions and causing the activations to become saturated.

6 CONCLUSION

The SAP approach guards networks against adversarial examples without requiring any additional training. We showed that in the adversarial setting, applying SAP to image classifiers improves both the accuracy and calibration. Notably, combining SAP with adversarial training yields additive benefits. Additional experiments show that SAP can also be effective against adversarial examples in reinforcement learning.